THE ROLE OF REACTIVE OXYGEN SPECIES FOLLOWING UV-B IRRADIATION

Redeat Tibebu, McNair Scholar

Gary Gardner Ph.D. and Doug Brinkman B.S., Department of Horticultural Science, University of Minnesota

Abstract

Ultraviolet (UV) radiation is a constituent of sunlight that influences plant morphology and growth. Due to the depletion of the ozone layer, the level of Ultraviolet-B (UV-B) with a wavelength range of 280–320 nm, has been increasing on the earth's surface. UV-B has been shown to induce photomorphogenic responses and cause damage to biomolecules in plants. However, the signaling molecules involved in the photomorphogenic response are unknown. Previous research on other stresses such as drought and heat have identified reactive oxygen species (ROS), toxic byproducts of cellular respiration, as signaling molecules in plants. This research examined the role of ROS as signaling molecule in *Arabidopsis thaliana* seedlings exposed to UV-B. Mutant lines for genes responsible for ROS as well as chemicals that interfere with ROS accumulation were used along with UV-B light. The results obtained suggest that ROS may play a signaling role in plants exposed to UV-B light.

Introduction

Effect of UV-B on Plant Growth

•UV-B (280–320 nm) reduces hypocotyl elongation in growing *Arabidopsis thaliana* seedlings (Gardner *et al*. 2009) and causes damage to biomolecules such as DNA, RNA and protein (Biever *et al*. 2014).

•The basis of these responses has not been well studied, and may be considered as a general reaction to stress (Borsche´ *et al*. 2003).

Role of Reactive Oxygen Species

• ROS play an important signaling role in plants controlling processes such as growth, development, response to biotic and abiotic environmental stimuli, and programmed cell death (American Society of Plant Biologists, 2006).

• During drought, heat, wounding and high-light stress, ROS accumulates at the site of stress (local) and away from the site of stress. This results in systemic signaling (Miller *et al*. 2009).

Focus of Study and Implications

• Examine the role of ROS in etiolated *A. thaliana* seedlings exposed to UV-B by using the NADPH flavin-oxidase inhibitor diphenylene iodonium (DPI) and the mutant line, ascorbate peroxidase (*apx2-1*).

• The results will hopefully determine whether ROS has a role in response signaling following UV-B exposure.

Materials

Mutant *A. thaliana* seeds*: apx 2-1*, was provided by Professor Ron Mittler at the University of North Texas. *xpf-3* was provided as described in et al. (Biever, 2014). Columbia-0 (Col-0) ecotype was purchased from Lehle Seeds (Round Rock, Texas, USA), and DPI was purchased from Sigma Aldrich co. (St. Louis, MO, USA).

UV light sources

UV light sources utilized are as described in Gardner et al. (2009). Broad-band UV-B light (FS40-T12-UVB-BP fluorescent tubes, UV Lighting Co., Brook Park, OH, USA) was used for fluence response analyses.

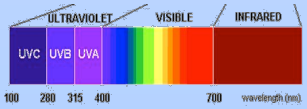
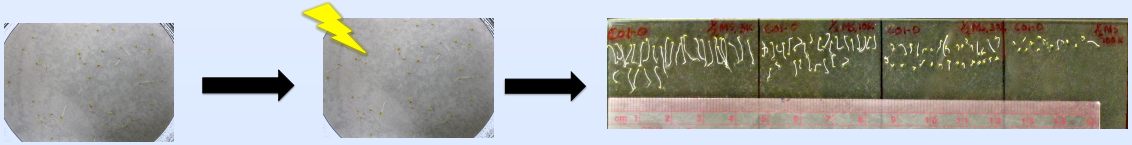


Fig.1: Spectral distribution of UV-B http://www.homesolariums.com.au/

Methods



1) 20-30 seeds of *A. thaliana* were planted on filter paper in a 10 cm Petri dish containing 0.7 ml ½ strength MS and 100 mm GA_4 or $GA_{4,7}$

2) When the radicles emerged (2-3 days after planting), DPI was applied (for DPI+UV-B test only) and each plate was irradiated with total fluences of $3x10^3$, 10^4, $3x10^4$, or 10^5 µmol m^{-2}.

3) After irradiation, plants were then returned to the dark for another 2–3 days of growth. Dark controls (plants not exposed to UV irradiation) were grown for all experiments. Seedlings were then transferred to a glass plate and photographed. Hypocotyl length was measured with the NIH image analysis program, ImageJ (http://rsb.info.nih.gov/ij/).

Results: Inhibition of Hypocotyl Elongation by DPI

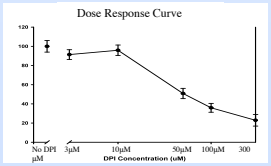


Fig. 4. Etiolated 2-3 day old *Arabidopsis* seedlings were treated with a range of DPI concentrations. Height of hypocotyls was measured 2 days after irradiation.

• Application of 10 µM DPI did not affect hypocotyl growth (Fig. 4).

• Hypocotyl growth was decreased by ~50 % by addition of 50 µM DPI.

Results: Inhibition of Hypocotyl Elongation by U-VB

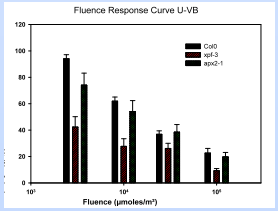


Fig.5a: Height of seedlings as a percent of the dark control 2 days after UV-B treatment. Etiolated 2-3 day old *Arabidopsis* seedlings were exposed to a range of UV-B fluences and hypocotyls were measured 2 days after irradiation.

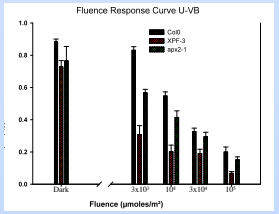


Fig.5b: Height of seedlings 2 days after UV-B irradiation.

• Fluence response curve shows the mutant line *xpf-3* to be more sensitive to UV-B than the wild type Col-0. (Fig. 5) This is consistent with previous results (Biever *et al*. 2014)

•Mutant line *apx2-1* is more sensitive to UV-B irradiation than the wild type at fluences of 3,000 and 10,000 µmoles m^{-2} of UV-B.

•Mutant *apx2-1* acts similar to the wild type Col-0 after 30,000 and 100,000 µmoles m^{-2} of UV-B

Results: Effect of DPI + UV-B Irradiation

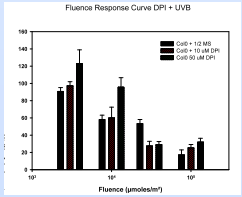


Fig. 6a.: Height of seedlings as a percent of the dark control 2 days after DPI and UV-B treatment.

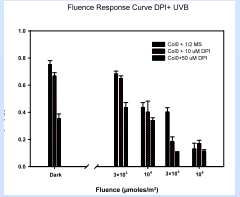


Fig.6b: Height of seedlings 2 days after DPI and UV-B treatment.

• 10 µM and 50 µM DPI were applied to Col-0 seedlings prior to UV-B exposure.

• The results show application of both 10 µM and 50 µM DPI altered the seedlings' sensitivity to UV-B (Fig. 6a & 6b).

• Seedlings treated with 10 µM DPI had increased sensitivity to 30,000 µmol m^{-2} UV-B.

• Seedlings treated with 50 µM DPI had decreased sensitivity to UV-B at 3,000 µmol m^{-2} and 10,000 and increased sensitivity at 30,000 µmol m^{-2}.

Discussion and Future Directions

• Mutant line *apx2-1*'s increased sensitivity to 3,000 and 10,000 µmol m^{-2} UV-B indicates that ascorbate peroxidase might be involved in the response of plants to UV-B.

• 10 µM DPI alone does not inhibit hypocotyl elongation

• Seedlings treated with 10 µM DPI had increased sensitivity to 30, 000 µmol m^{-2} UV-B, indicating that NADPH flavin-oxidase may be involved in the response to UV-B.

• Future studies will repeat experiments on *apx2-1* as well as other mutations lacking enzymes involved in ROS metabolism.

References

1. Biever, J., Brinkman D., Gardner G. 2014, UV-B inhibition of hypocotyl growth in etiolated *Arabidopsis thaliana* seedlings is a consequence of cell cycle arrest initiated by photodimer accumulation. Experimental Botany. 65(11): 2949-2961
2. Gardner, G., C. Lin, E.M. Tobin, H. Loehrer, and D. Brinkman. 2009. Photobiological properties of the inhibition of etiolated Arabidopsis seedling growth by ultraviolet-B irradiation. Plant, Cell Environ. 32(11): 1573-1583
3. Borche' M., Schuler M.A., Kalbina I., Conner L. & Stride A (2002) Gene regulation by low level of UV-B radiation: identification by DNA array analysis. *Photochemical & photobiological Sciences* (1): 656-664.
4. Miller, G., K. Schlauch, R. Tam, D. Cortes, M.A. Torres, V. Shulaev, J.L. Dangl, and R. Mittler. 2009. The Plant NADPH Oxidase RBOHD Mediates Rapid Systemic Signaling in Response to Diverse Stimuli. Sci. Signal. 2(84):ra45.

Acknowledgments

I would like to thank Dr. Gary Gardner for his willingness to involve me in his research and for being a great mentor. I would also like to thank Doug Brinkman for all his help throughout my research. In addition the Department of Horticultural Science and the McNair staff have been very supportive and helpful.